

Indian Institute of Technology Roorkee



Summer Internship

Estimation of Traffic Flow Using Google API

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Request for Verification of Research Internship

2 messages

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Tue, 9 Sept 2025 at 22:41

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Respected Sir,

I am writing to kindly request your verification of my Research Internship carried out under your guidance from May 2025 – October 2025. During this period, I (Hansika Soni), along with Maddipati Sathvika, have been working on the project titled "Estimation of Traffic Flow Data Using Google API."

Kindly confirm this internship for official purposes.

Best regards,
Hansika Soni
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Satish Chandra <satish.chandra@ce.iitr.ac.in>

Wed, 10 Sept 2025 at 10:59

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Yes, I verify it.

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Maddipati Sathvika

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1. Introduction

Traffic flow estimation plays a vital role in urban planning, congestion management, and roadway design. Traditional approaches to measuring flow, such as field surveys and sensor-based systems, are expensive, labour-intensive, and difficult to scale. With the emergence of real-time digital data sources like the Google Distance Matrix API, traffic monitoring can be performed more efficiently using stream speed as a proxy for flow.

This study explores multiple modelling approaches—ranging from classical linear regression and density-based methods to advanced machine learning techniques such as Random Forest, Gradient Boosting, KNN, and regression variants (Huber, Quantile). The primary objective is to predict vehicle volumes in three traffic categories (small, medium, and large vehicles) using stream speed data. The performance of these models is evaluated to identify the most suitable approaches for accurate, scalable, and cost-effective traffic flow estimation.

2. Problem statement

The problem addressed in this project is the estimation of traffic flow using stream speed data obtained from the Google Distance Matrix API. Traditional traffic monitoring methods, such as manual counting or sensor-based systems, are often costly, labour-intensive, and limited in their ability to capture data continuously. In contrast, stream speed offers a readily available and scalable metric that can be used to infer traffic patterns. The objective of this project is to predict the number of different vehicle types—such as cars, bikes, trucks, and buses—present on a road segment at any given time using real-time stream speed data. This provides an efficient and practical alternative for traffic monitoring and management without the need for extensive physical infrastructure.

To simplify the prediction process, all different vehicle types are grouped into three broader categories based on their size and impact on traffic flow:

- **Small vehicles** → Two-wheelers such as bikes and scooters.
- **Medium vehicles** → Three-wheelers and light motor vehicles (LMVs) such as cars and vans.
- **Large vehicles** → Heavy vehicles such as buses and trucks.

This categorization reduces complexity in modelling while still capturing the essential differences in traffic composition, enabling more accurate and practical traffic flow estimation.

3. Origin of the Idea

In traffic engineering and transportation research, the estimation of basic flow parameters—**flow, speed, and density**—is essential for a wide range of traffic-related tasks such as performance evaluation, congestion analysis, capacity estimation, and traffic control. Traditionally, the collection of such data relies on **field surveys** through methods like video recordings, roadside counts, or installation of physical sensors. While effective, these techniques are **labour-intensive, time-consuming, and cost-intensive**, often requiring significant manpower and resources to capture data continuously and across multiple locations.

Recognizing these limitations, the idea for this research was motivated by the possibility of leveraging **digital data sources**, specifically the **Google Distance Matrix API**, to obtain real-time traffic information. The API provides live travel times and speeds, which can potentially be utilized to **estimate traffic volume indirectly**. If a robust model can be developed to infer traffic volume from **stream speed data alone**, it would eliminate the need for continuous field-based data collection.

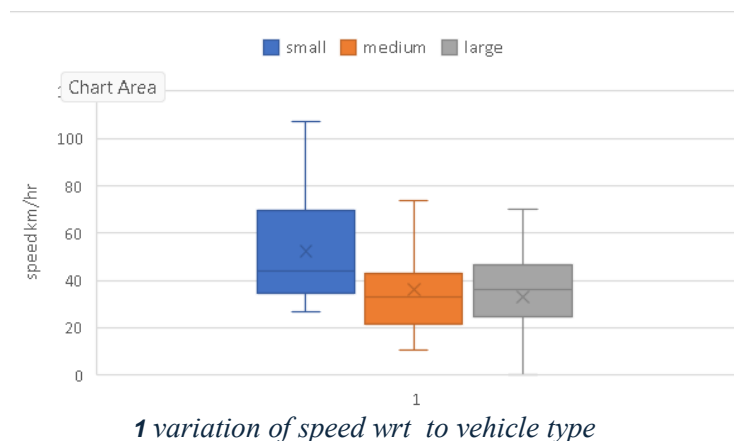
This approach, though still in its **early research stage**, presents a promising opportunity to simplify traffic data collection. It allows researchers and practitioners to gather reliable traffic flow parameters **remotely, efficiently, and at minimal cost**, ultimately making traffic monitoring more scalable and accessible.

4. Data

Training Dataset

The training dataset used in this project was compiled from four different locations to capture variations in traffic conditions. It contains two key types of information:

- **Stream speed parameters** such as small vehicle velocity (V_s), medium vehicle velocity (V_m), and large vehicle velocity (V_L).
- **Vehicle counts** classified into small (N_s), medium (N_m), large (N_L), and the total number of vehicles (N_t).



This dataset forms the foundation of the predictive model. By analysing the relationship between stream speed values and vehicle counts across multiple locations, the model is trained to recognize patterns that reflect how different traffic compositions influence speed conditions. The use of data from multiple sites enhances the robustness and applicability of the model in real-world scenarios.

Testing dataset

Google Distance Matrix API and Its Role in Traffic Estimation

What is the Distance Matrix API?

- A service provided by Google that returns **distance and travel time** for origin–destination pairs.
- Accounts for **real-time traffic conditions**, road network geometry, and travel mode.
- Commonly used in navigation apps, logistics, and route optimization.

Key Uses of the API

- **Travel Time Estimation** → Provides dynamic travel times under current traffic conditions.
- **Speed Derivation** → From distance and travel time, the average **stream speed** can be calculated.
- **Traffic-Aware Routing** → Helps in route planning by considering congestion and delays.
- **Scalable Data Access** → Can gather information across multiple locations simultaneously without physical deployment.

Relevance in This Research

- **Cost-Efficient Alternative** → Eliminates the need for field-based surveys such as videography, manual counts, or sensors.
- **Remote Monitoring** → Enables researchers to gather live traffic speed data from anywhere, “sitting at home.”
- **Foundation for Estimation Models** → Stream speed data serves as the **input** for models that estimate traffic volumes and other flow parameters.
- **Towards Simplified Traffic Analysis** → Offers a pathway to derive essential flow parameters (volume, speed, density) in a more automated and scalable manner.

For the testing phase, live data was collected to evaluate the trained model. The **stream speed values** were obtained in real time from the **Google Distance Matrix API**, while **video footage was recorded simultaneously from a foot-over bridge** overlooking the road section. This ensured that speed data and actual traffic flow observations were aligned in both time and location, making the dataset reliable for validation.

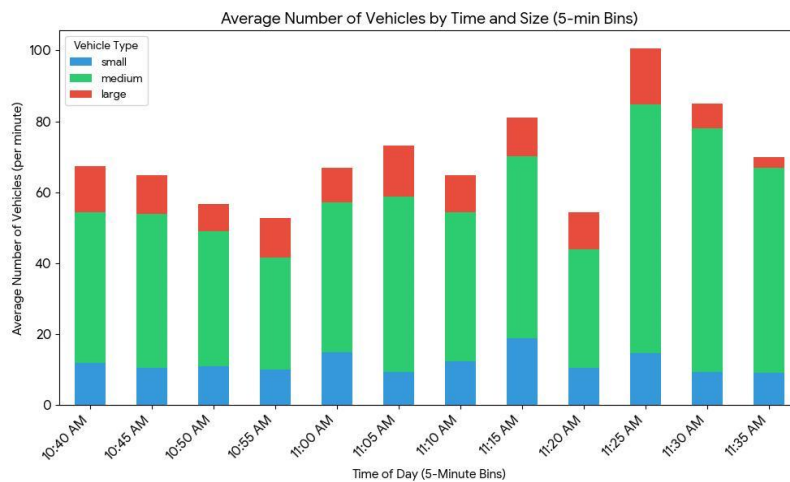


3 Camera setup for traffic data recording.



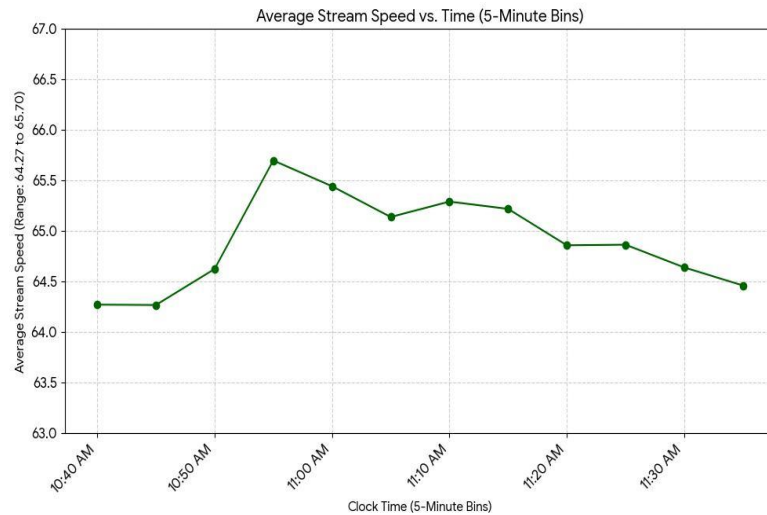
2 Defined trap region on the roadway for vehicle detection

The vehicle counts from the recorded video were extracted using the **YOLOv8s object detection model**. YOLOv8s (You Only Look Once, version 8 - small) works by dividing each video frame into a grid and predicting bounding boxes and class probabilities for objects within those grids. It is optimized for real-time detection, allowing it to identify vehicles such as cars, bikes, buses, and trucks with high accuracy and speed. By processing the video frames, YOLOv8s provided the number and type of vehicles present, which served as the ground truth for evaluating the accuracy of the model's predictions.



4 Extracted no. of vehicles vs time

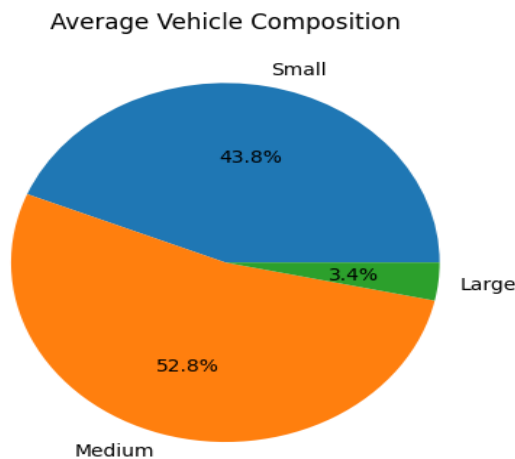
This dual collection ensured that the model’s predictions could be directly validated against observed traffic flow. The live stream speed acted as the input for prediction, whereas the vehicle counts extracted from the videos served as the benchmark for comparison. This approach allowed for reliable testing of how accurately the model estimates small, medium, and large vehicle categories under real-world conditions.



5 Live stream speed vs time

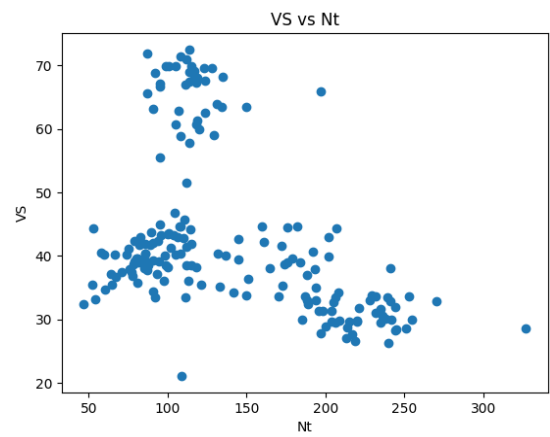
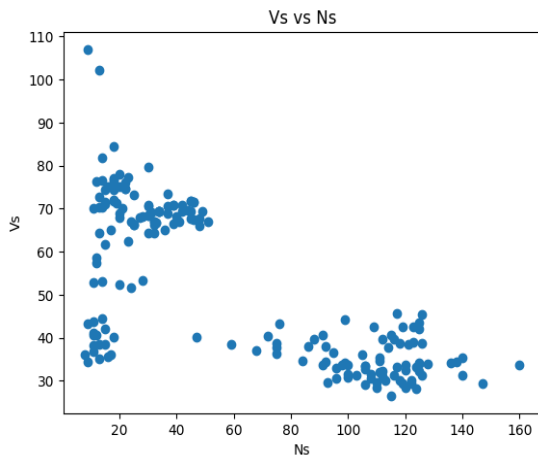
5. Exploratory Data Analysis

- Average Composition of vehicles



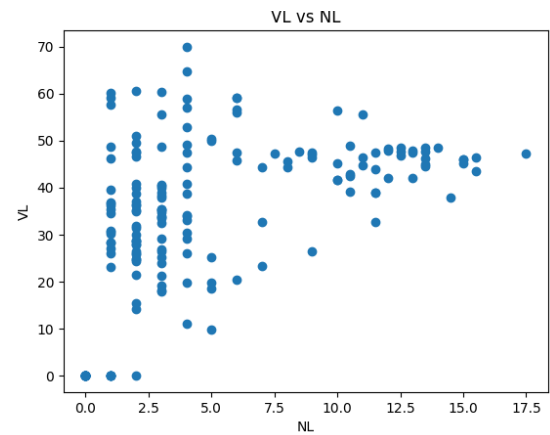
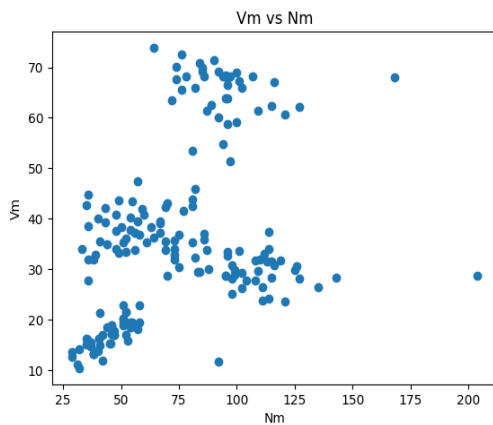
6 Vehicle composition in training dataset

- **Volume vs speed variations**



10 medium vehicle velocity vs volume

9 small vehicle velocity vs volume

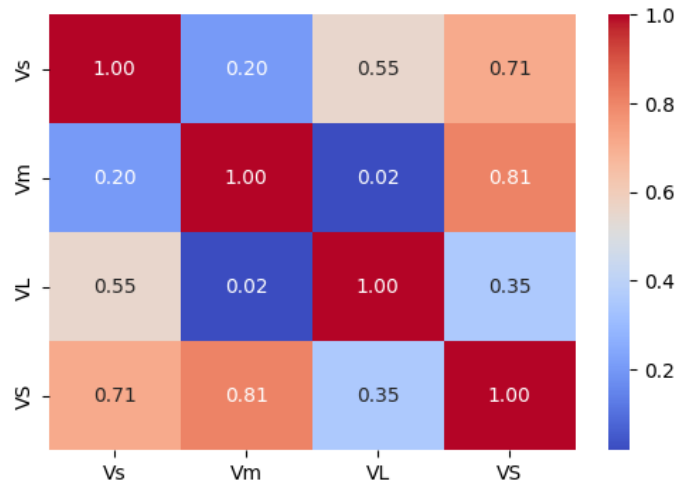


7 Stream speed vs total volume

8 large vehicle velocity vs volume

The scatter plots of velocity against the number of vehicles for each category (small, medium, and large) do not reveal any clear or consistent trend. The points appear randomly distributed, with no fixed mathematical relationship or observable order between vehicle count and speed. This indicates that, in the given dataset, the velocity of vehicles within each category is not directly governed by their volume. Instead, the traffic flow seems to behave in a stochastic manner, likely influenced by external factors such as road geometry, signal timing, lane discipline, and driver behaviour, which are not explicitly captured in the dataset.

- **Correlation between stream speed and individual category speeds**



11 correlation heatmap between speeds

- VS vs Vm → 0.81 (Strong Positive Correlation)**
 - Stream speed (VS) is **most strongly influenced by medium vehicle speed**.
 - This indicates that **medium vehicles contribute significantly** to overall traffic flow dynamics.
- VS vs Vs → 0.71 (Moderately Strong Correlation)**
 - Small vehicle speed also has a **substantial effect** on stream speed.
 - Suggests that small vehicles, being large in number, **pull VS upward** when they move faster.
- VS vs VL → 0.35 (Weak Positive Correlation)**
 - Large vehicles have the **least influence** on stream speed.
 - This is expected, as heavy vehicles are fewer and slower, so they only marginally affect VS.

6. Methodology

To estimate traffic volumes from stream speed data, a variety of statistical and machine learning models were applied. Each model was chosen for its unique strengths in handling different data characteristics, and its performance was evaluated across small, medium, and large vehicle categories.

- **K-Nearest Neighbours (KNN)**

KNN is a non-parametric, instance-based learning algorithm that predicts outcomes by averaging the values of the nearest neighbours in the feature space. It makes no assumptions about the underlying data distribution.

In this work, KNN was applied to identify local traffic patterns by comparing current stream speed conditions with similar historical cases. It proved particularly effective for medium and

large vehicle categories, where flow patterns followed more consistent behaviours. This highlighted the usefulness of local similarity-based approaches in traffic estimation.

- **Gradient Boosting**

Gradient Boosting is an ensemble method where trees are built sequentially, with each new tree correcting the errors of the previous ones. This allows the model to capture complex, non-linear relationships between variables.

Here, Gradient Boosting was used to learn the residual patterns between stream speed and vehicle volumes. It provided balanced and stable predictions, showing strength in modelling non-linear traffic behaviours. Compared to Random Forest, it adapted better to gradual error corrections but was more sensitive to fluctuations in the data.

- **Random Forest**

Random Forest is an ensemble method that constructs multiple decision trees on randomly selected subsets of the data and aggregates their predictions. This reduces overfitting and improves generalization.

In the project, Random Forest was applied as a multi-output regression model to estimate small, medium, and large vehicle counts simultaneously from stream speed. It performed consistently across all categories and was particularly effective in handling the variability of small vehicle flows. Its robustness made it one of the most reliable models tested.

- **Support Vector Regression (SVR)**

SVR is a machine learning method that maps data into a higher-dimensional space using kernel functions and fits a regression line within a defined margin of tolerance. It is effective in capturing non-linear patterns and is robust to outliers.

For this problem, SVR was trained using stream speed values to estimate vehicle volumes across categories. It showed stronger performance for large vehicles, where flow patterns were more stable, but struggled with highly variable flows like small and medium vehicles. This indicated that SVR can model structured traffic patterns but is less suitable when randomness dominates.

- **Hybrid Density Approach**

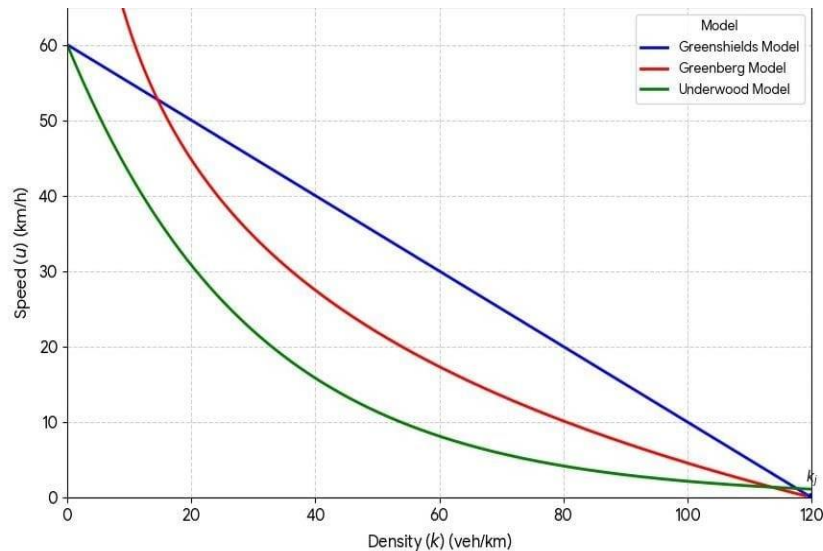
The Hybrid Density approach was developed as a **custom traffic modelling framework**, combining classical traffic flow theories with modern data-driven learning. Unlike predefined machine learning algorithms, this approach was specifically designed to exploit the physical relationships between speed, density, and flow.

Classical models such as **Greenshields, Greenberg, and Underwood** each describe the speed–density relationship in traffic, but none alone can fully represent all operating conditions. To overcome this limitation, a **piecewise hybrid model** was constructed by blending these theories in different speed ranges. The hybrid formulation ensures that:

Greenshields [9]	$v = v_f \left(1 - \frac{k}{k_j}\right)$	v_f, k_j
Greenberg [10]	$v = v_m \ln \left(\frac{k_j}{k}\right)$	v_m, k_j
Underwood [11]	$v = v_f e^{-\frac{k}{k_m}}$	v_f, k_m

12 Formulae for Model (Source: Chrome)

- At **low speeds**, traffic behaviour is better captured by Greenberg's logarithmic form.
- At **moderate speeds**, Underwood's exponential relationship provides a realistic fit.
- At **high speeds**, Greenshields' linear assumption becomes more dominant.



13 Results of Hybrid Density Approach

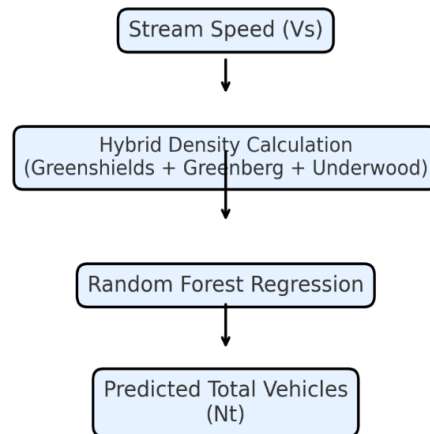
This blending produced a **composite density measure**, denoted as hybrid, which adapts to varying traffic states more effectively than any single model.

Once density was estimated from stream speed using this hybrid formulation, it was provided as input to a **Random Forest regression model** to predict vehicle counts. Two configurations were tested:

1. **Density-only model**, where predictions were based solely on the calculated density.
2. **Density + speed model**, where both density and stream speed were included as inputs.

Interestingly, both approaches gave identical results, indicating that the hybrid density measure already **encapsulates the essential traffic information** contained in stream speed. This shows the strength of the method: it bridges **theory-driven traffic flow models** with **data-driven learning**, creating a robust and interpretable framework for vehicle count estimation.

Hybrid Density → Total Vehicle Estimation (Nt)



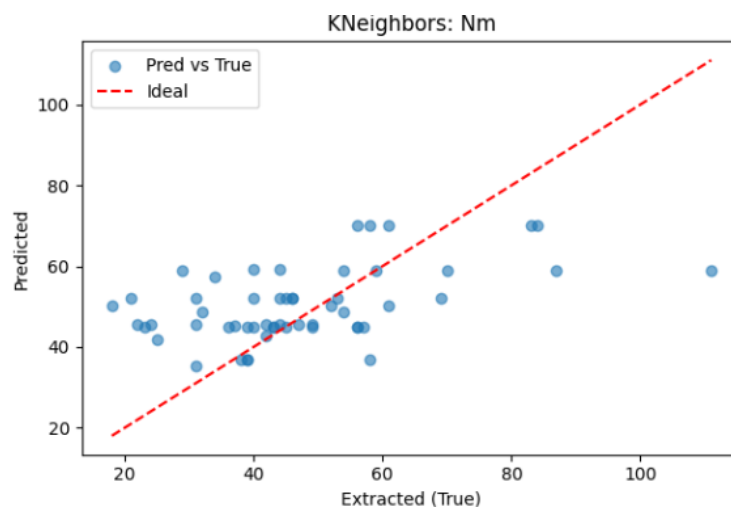
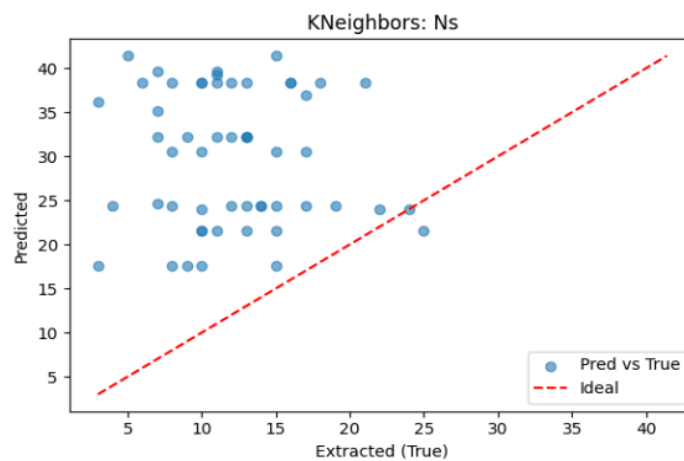
7. Results

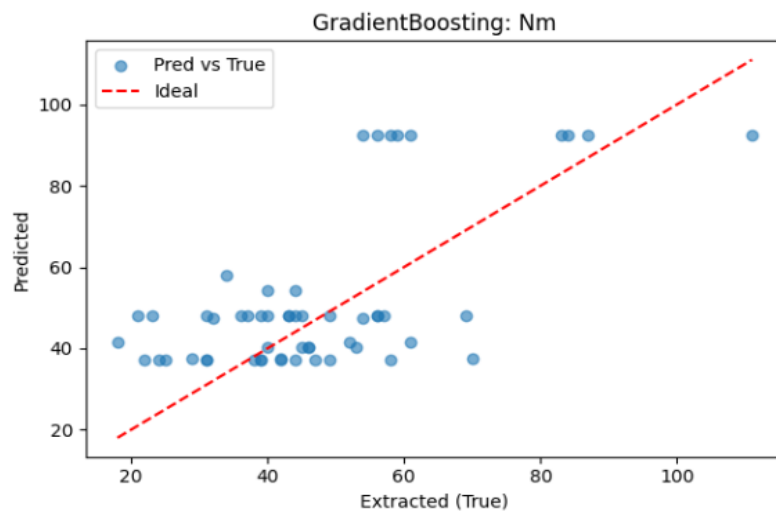
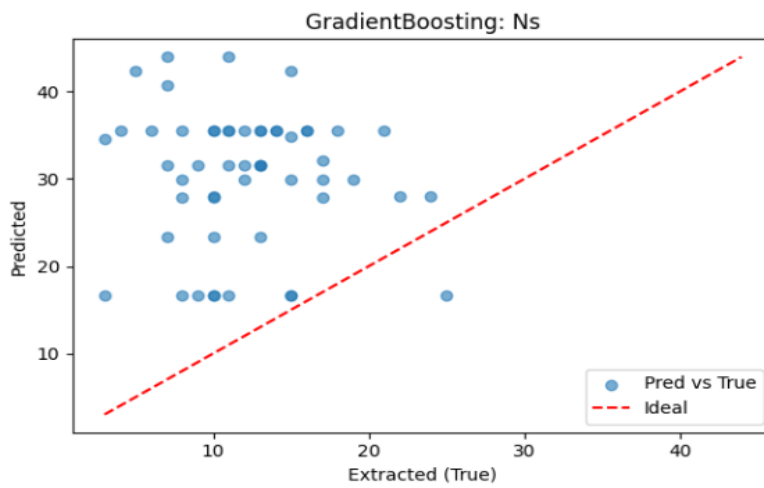
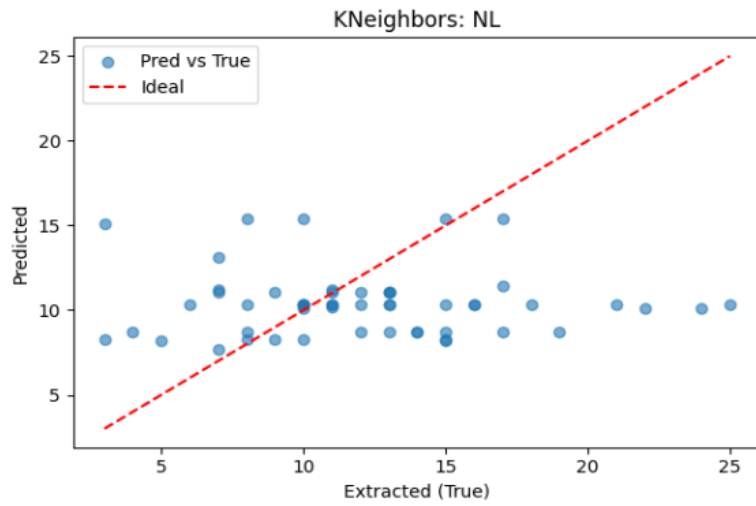
RMSE:

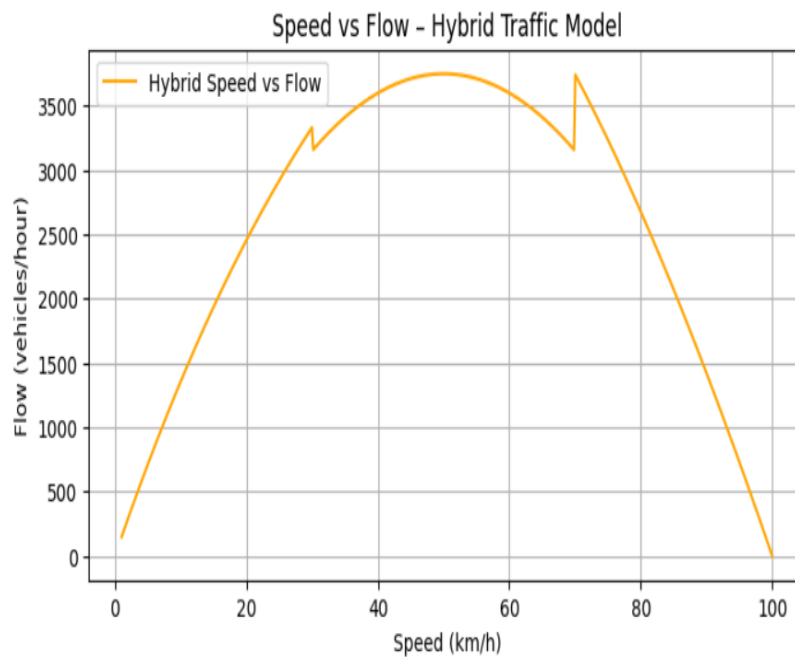
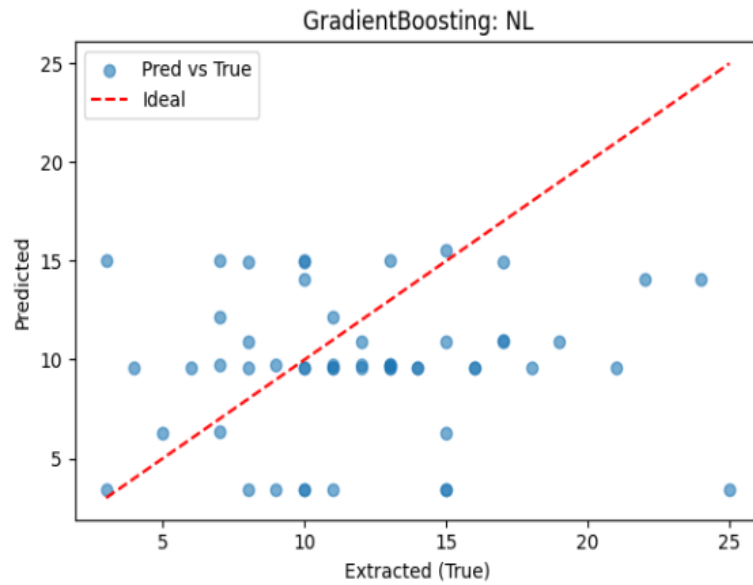
Model/Approach	Small Vehicles (Ns)	Medium vehicles (Nm)	Large Vehicles (NL)	Combined/Total
KNN	20.0552	15.5867	5.550108	15.01066
Gradient Boosting	20.54885	16.32392	6.304705	15.58284
Random Forest	21.0483	16.32392	6.304705	15.69147
SVR	22.95631	19.13519	4.950667	17.48959
Linear Regression	27.64483	31.79489	7.331862	24.69081
Hybrid Density				31.12

MAE:

Model/Approach	Small Vehicles (Ns)	Medium vehicles (Nm)	Large Vehicles (NL)
KNN	17.91481	11.72222	4.111111
Gradient Boosting	18.62232	12.96025	4.94045
Random Forest	18.65228	13.21208	4.819353
SVR	22.37018	15.25634	3.771018
Linear Regression	27.05138	29.21132	5.924308







8. Inference:

- **KNN achieved the lowest combined SE**, making it the best overall model.
- **Random Forest produced the best results for small vehicles**, showing strong stability in high-variability flows.
- **KNN performed best for medium and large vehicles**, capturing local traffic patterns effectively.
- **SVR achieved the lowest error for large vehicles** but was weaker for small and medium categories.
- The **Hybrid Density approach** performed reasonably well for total vehicle prediction but was less accurate than machine learning models for category-wise estimation.

9. Conclusion:

ML models like KNN, Random Forest, and SVR successfully captured traffic flow patterns but struggled with inconsistent traffic behaviour across different vehicle categories and road sections, leading to variable accuracy.

The main challenge came from the highly fluctuating flow of small and medium vehicles compared to the more stable patterns of large vehicles. This variation across locations reduced the predictive power of all models.

Although the hybrid density model provides a logical and theory-backed approach, it can only predict total traffic flow. Predicting individual vehicle categories requires location-wise category share data, which was not available in this study.

While still in the research stage, current results indicate that stream speed alone is insufficient for accurately predicting individual category flows; additional features or data sources are needed for better model performance.